

机器灌注条件下细胞外囊泡调控移植器官缺血再灌注损伤及早期排斥风险：机制、标志物与转化应用

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Abstract

缺血再灌注损伤 (ischemia-reperfusion injury, IRI) 贯穿实体器官移植的供体获取、保存、植入和早期免疫激活过程,是多器官移植共同面对的病理枢纽。近五年,低温含氧机器灌注、常温机器灌注和控制性复温逐步把器官保存从静态冷保存推向可实时监测、可干预的离体平台;细胞外囊泡 (extracellular vesicles, EVs) 也从旁观性分泌产物,被重新理解为损伤信号、免疫调控载体和潜在治疗递送系统。现有证据提示, EV cargo 中的 miRNA、蛋白、脂质和线粒体相关成分能够影响内皮糖萼、补体、炎症小体、巨噬细胞极化、铁死亡/焦亡和内皮-实质细胞串扰;机器灌注则为 EV 的器官靶向递送、灌注液液体活检、短时药效评价和可逆性修复提供了难得的操作窗口。本文在既有移植 IRI、机器灌注和 EV 文献基础上,围绕 HOPE、NMP、亚常温灌注和控制性复温的模式差异,梳理 EV 作为病理信号、标志物和治疗载体的证据链。转化难点仍集中在 EV 来源、cargo 稳定性、剂量单位、纯度、效价、器官内分布、长期免疫后果和监管质量标准。未来 3-5 年更值得推进的工作,是把“EV 治疗”从宽泛的抗炎叙事,落实到灌注平台上的可计量药代、可解释标志物、可放行产品和小样本自适应临床研究。

1 引言

IRI 并不是单纯的氧化应激事件,而是冷缺血、复温、线粒体损伤、内皮屏障破坏、补体启动、固有免疫细胞募集和后续适应性免疫放大的连续过程。肝、肾、心、肺等器官在解剖和代谢上不同,但早期损伤均可通过 DAMPs、线粒体 DNA、内皮糖萼脱落、NLRP3 炎症小体和微循环障碍汇入相似的免疫激活网络 [1-4]。临床上,这一网络会转化为移植物功能延迟、原发无功能、胆道并发症、急性排斥易感性以及慢性移植物损伤的底层土壤。因而,围绕 IRI 的治疗不应只追求单个炎症因子的短时下降,更应关注能否在植入前改变器官的代谢储备、内皮完整性和免疫呈递阈值。

机器灌注的价值首先体现在干预时点前移。静态冷保存使临床团队主要在植入后处理损伤,而机器灌注允许在移植前评估代谢恢复、微循环阻力、胆汁/尿液生成、乳酸清除、酶学释放和灌注液分子标志物 [5-7]。器官保存因此不再只是“运输技术”,也成为短时体外药理学平台。EV 的兴起正好进入这一场景: EV 可以作为损伤读数,也可以作为治疗载体;既可来源于供体器官、免疫细胞、间充质 stromal cells、尿液祖细胞、脂肪来源 stromal cells 或器官局部细胞,也可通过工程化方式装载抗炎、抗氧化、抗铁死亡或免疫调节 cargo [8-10]。

本文关注的核心问题不是“EV 是否有抗炎作用”这一泛化命题,而是机器灌注怎样改变 EV 研究的证据形态。离体灌注把供体器官从受体全身环境中暂时分离出来,降低了系统性免疫、药物分布和肝肾清除对结果解释的干扰;同时,灌注液可重复采样,使 EV 浓度、cargo、炎症介质、代谢物和器官功能读数可以在同一时间轴上观察。因此, EV 在这一场景中至少有三种角色:第一,作为供体器官损伤和免疫原性的输出信号;第二,作为灌注期修复药物或递送载体;第三,作为评价机器灌注是否真正改变器官状态的伴随诊断工具。这三种角色相互重叠,也正是后续转化研究最容易混淆的地方。

2 IRI 的免疫机制：从线粒体失衡到排斥风险

IRI 的核心机制是能量失败后再供氧所触发的非线性放大。冷缺血阶段 ATP 耗竭、离子泵失衡和线粒体电子传递链损伤已经埋下伏笔；复温和再氧合阶段，ROS 暴发、钙超载、膜通透性转换孔开放和线粒体 DAMPs 释放共同激活内皮、Kupffer 细胞、肾小管细胞和组织驻留免疫细胞 [11–13]。这一步的关键不只是 ROS 数量增加，而是线粒体从能量器官转变为免疫刺激源：mtDNA、心磷脂、N-formyl peptide 和受损线粒体片段可被模式识别受体、补体和中性粒细胞识别，推动内皮黏附分子上调、微血栓形成和组织水肿。补体是这一过程中特别值得靶向的层级，因为它既连接先天免疫，也能通过 C3a/C5a、膜攻击复合物和凝血串扰放大微血管损伤 [2]。

细胞死亡方式的多样化解释了为什么单一抗氧化药物很难稳定改善移植结局。凋亡、坏死性凋亡、焦亡、铁死亡、线粒体通透性转换相关坏死和自噬/线粒体自噬失衡，可在不同器官和不同缺血强度下并行存在 [4, 13]。例如，脂肪变性肝更容易发生脂质过氧化和微循环障碍，DCD 心脏更突出 warm ischemia 后的线粒体肿胀和收缩功能恢复问题，扩展标准供肾则常表现为肾小管上皮损伤、局部补体生成和延迟移植物功能。机器灌注的意义在于，它可以把这些机制拆成可测量环节：低温阶段重建 ATP 和降低 succinate 负荷，复温阶段观察血管阻力和酶学释放，常温阶段评价胆汁、尿液、乳酸清除和功能储备。

EV 在这一网络中具有双重身份。损伤细胞释放的 EV 可携带氧化脂质、miRNA、线粒体片段和组织特异蛋白，成为“损伤正在发生”的信号；治疗性 EV 则可能通过转运抗凋亡 miRNA、免疫调节蛋白、抗氧化酶相关信号和线粒体稳态相关成分，削弱炎症与细胞死亡 [10, 14, 15]。病理性 EV 并非理论假设；心肌 IRI 中释放的心源性 EV 可呈现促炎特征并加重损伤，提示 EV 既可能是修复载体，也可能是 DAMPs 传播工具 [16]。问题在于这两类 EV 在粒径、来源、cargo 和生物学效应上并不天然可分。若没有来源追踪和效价定义，EV 研究容易把“检测到囊泡变化”误读为“囊泡驱动病理”。

从排斥风险看，IRI 与免疫原性之间存在明确的生物学连接。内皮糖萼脱落会暴露黏附分子和凝血表面，补体裂解产物促进抗原呈递细胞活化，坏死/焦亡释放的核酸和蛋白增强局部炎症，组织驻留淋巴细胞也可在机器灌注过程中被释放到灌注液中 [3, 17, 18]。因此，EV 研究若只观察移植后早期肌酐或转氨酶变化，可能低估其长期免疫效应；更合理的设计应同时记录急性损伤、免疫细胞迁移、供体特异性抗体、分子排斥标志和慢性纤维化端点。

3 机器灌注作为 EV 递送窗口

机器灌注并不是单一技术，而是一组温度、氧合、流量、压力、灌注液成分和时序选择的组合。HOPE 和其他低温含氧灌注主要利用低温降低代谢需求，同时通过氧合维持线粒体电子传递链的最低限度恢复，减少再灌注早期 ROS 暴发；NMP 则把器官置于接近生理温度的代谢状态，允许功能评估、药物代谢和更复杂的细胞反应；亚常温灌注和控制性复温位于两者之间，试图避免从冰冷保存直接跳到常温再氧合所带来的温度冲击 [6, 19, 20]。这些模式对 EV 治疗的含义完全不同：低温有利于器官保护和物流，但可能限制主动内吞、胞内转运和 cargo 释放；常温有利于药效观察和细胞摄取，却增加 EV 在含血灌注液中的蛋白冠形成、吞噬清除和促凝风险；控制性复温则可能成为先清除有害 EV、再递送修复性 EV 的折中窗口。

机器灌注的最大优势是把治疗递送限制在器官层面。对于 MSC-EV、尿液祖细胞来源 EV 或工程化纳米囊泡，离体灌注可以减少全身给药的稀释、免疫副作用和非靶器官分布，同时允许在植入前洗脱未结合成分 [21–23]。在扩展标准供肾和 DCD 心脏模型中，MSC-EV 通过 HOPE 或相关灌注策略递送后显示出抗炎、抗凋亡和改善微循环的信号；猪 DCD 心脏模型中的初步研究则支持 EV 对心肌 IRI 具有保护潜力 [24]。尿液祖细胞来源 EV 在猪肾 HMP 后接 NMP 的模型中显示出可行性和安全性，联合 HMP/NMP 给药组出现细胞损伤标志物下降、组织完整性改善以及代谢和细胞因子谱改变，但短时肌酐清除和 NGAL 等功能指标并未同步显著改善 [23]。这一结果对转化很重要：EV 可能先改变损伤生物学，再转化为可见功能获益；早期试验若只设置粗粒度临床功能端点，可能无法捕捉真实药效。

不过，现有证据仍主要来自小动物、大动物和离体概念验证。EV 在灌注液中的稳定性、血管内皮摄取、实质细胞渗透深度、胆道/肾小管分布，以及不同温度、氧合状态和剪切力下的 cargo 保留，仍缺少可比较的数据。常温机器灌注更接近生理代谢，适合评价即时药效和组织摄取；低温含氧灌注更有利于线粒体恢复和 ROS 预处理，但 EV 内吞和细胞代谢反应可能受低温限制。因此，EV 递送必须放在具体灌注模式中讨论。

3.1 EV 来源与 cargo：从细胞类型到效价定义

EV 来源决定了其免疫语义。MSC-EV 是当前最常被讨论的治疗产品，优势在于免疫调节、旁分泌修复和较成熟的细胞制备经验；但 MSC 组织来源、供体年龄、培养密度、血清替代物、低氧预处理和传代次数都会改变 EV cargo [25–27]。脂肪来源 stromal cell、骨髓 MSC、脐带 MSC、尿液祖细胞和肝源性 stromal-like cells 释放的 EV 不应被视为同一产品的不同批次，而更接近不同药物类别。器官局部细胞来源 EV 可能具有更好的归巢或组织语境，但制备规模、供体差异和病理 cargo 污染更难控制。

cargo 层面，miRNA 是最常被赋予机制解释的成分，例如调控 ferroptosis、SLC7A11、HO-1 相关抗氧化轴、巨噬细胞极化或 Th17/Treg 平衡的 miRNA 组合 [28–31]。但 EV 的效应不能简化为单个 miRNA。膜蛋白决定细胞摄取和免疫识别，脂质组成影响补体/凝血反应，表面糖基化可能改变内皮黏附，残留蛋白和培养基污染则会造成假阳性药效。工程化 EV 或杂合纳米囊泡可以提高装载量和靶向性，但同时把监管属性从天然分泌产物推向复杂生物制品/递送系统 [8, 9, 32]。因此，灌注期 EV 研究需要从一开始就定义效价：是以内皮糖萼保护、补体沉积下降、线粒体膜电位恢复、炎症小体抑制、胆汁/尿液功能改善，还是以植入后 DGF/PGD 下降作为主要效应。

3.2 递送时序：HOPE、NMP 与控制性复温的不同任务

HOPE 适合把 EV 与线粒体保护、补体靶向或内皮保护策略结合。其优势是器官代谢需求低、灌注压力可控、未结合 EV 可在植入前冲洗；局限是低温下受体细胞主动摄取较弱，EV 更可能停留在血管腔、内皮表面或被灌注液成分吸附。NMP 适合观察 EV 是否改变乳酸清除、胆汁质量、肾血流、尿液生成、心肌酶释放或肺顺应性，也适合与 MSC 生物反应器、基因递送和代谢干预联合 [33–35]。控制性复温的价值在于把温度和氧合逐步推高，允许研究者区分“低温保护”、“复温损伤”和“常温修复”三个阶段。若将 EV 用于治疗，较合理的策略可能不是单次固定给药，而是按灌注阶段分层：低温/复温早期给予内皮和补体保护型 EV，常温评价阶段再给予促代谢

恢复或免疫调节型 EV，并通过灌注液读数决定是否追加剂量。

4 灌注液 EV 与分子标志物

EV 的另一条转化路径是标志物。灌注液比外周血更接近供体器官，背景噪音低，可连续采样，适合捕捉内皮、胆管、肾小管、心肌和免疫细胞释放的时间序列信号。近年研究开始把 EV、miRNA、代谢物、蛋白组和细胞游离 DNA 共同纳入器官质量评价框架 [36–38]。灌注液 EV/miRNA 被清除或富集后的变化，也提示“过滤有害囊泡”可能与“递送有益囊泡”形成互补策略 [39]。

临床可用的标志物不应停留在单个 miRNA 或单个炎症因子，而要回答三个问题：器官是否适合移植、是否需要灌注期修复、植入后早期是否需要强化监测。为此，灌注液 EV 指标需要与灌注动力学、组织病理、术后功能延迟、dd-cfDNA、供受体免疫风险和影像/病理端点耦合。仅有相关性不足以改变临床决策；前瞻性阈值、阴性预测价值和干预后的风险重分类证据同样关键。

灌注液 EV 标志物的优势来自采样位置，但其解释同样受灌注模式影响。HOPE 和 HMP 的低温、低代谢背景可能更适合捕捉缺血期积累的内皮/肾小管损伤 EV；NMP 中的 EV 谱则同时反映损伤释放、免疫细胞活化、血液成分加入、细胞摄取和再分泌。控制性复温研究显示，灌注液 EV 和 miRNA 可以被亲和吸附装置移除，这一方面支持 EV/miRNA 作为免疫原性 cargo 的可操作靶点，另一方面也提示标志物研究必须记录采样前是否存在滤过、吸附、换液或血液制品更换 [36, 39]。如果不控制这些工程变量，同一 EV 指标在不同中心之间很难比较。

更实用的标志物框架应包含三层。第一层是器官来源和细胞来源判定，例如肾小管、胆管、内皮、心肌或肺泡上皮相关蛋白/miRNA 组合，用于判断主要受损解剖单元。第二层是机制读数，包括补体、凝血、炎症小体、线粒体应激、铁死亡和糖萼脱落信号，用于决定是否需要灌注期干预。第三层是临床决策读数，即与移植后 DGF、PGD、胆道并发症、急性排斥或 dd-cfDNA 升高相关的预测模型 [18, 40, 41]。这三层不能互相替代：机制读数适合药效证明，临床决策读数适合器官取舍，而来源读数决定干预靶点。

技术上，灌注液 EV 检测至少要面对分离方法和归一化问题。超速离心、尺寸排阻、沉淀、免疫捕获和微流控方法会富集不同 EV 亚群，也会带入不同程度的脂蛋白、蛋白聚集体和血小板来源颗粒。灌注液的体积、血细胞含量、白蛋白/红细胞制品、肝素和吸附柱使用都会影响粒子计数。单纯以每毫升粒子数报告，不能区分器官释放增加、灌注液浓缩、细胞破裂或外源性 EV 残留。更稳妥的做法，是同时报告粒子数、蛋白量、特异 marker、采样时间、灌注液体积变化和与器官重量/灌注流量相关的归一化指标，并把 EV 数据与蛋白组、代谢组和组织病理共同解释 [38, 42]。

5 讨论：主要瓶颈

主要瓶颈有四类。EV 产品本身高度异质，MSC 来源、培养条件、缺氧预处理、纯化方法、粒径分布、蛋白污染、RNA cargo 和效价检测都会影响结果。剂量单位仍不统一，粒子数、蛋白量、细胞当量和功能效价之间缺少换算。安全性评估也不能只看急性毒性，还要覆盖促凝、促血管生成、免疫耐受过度、潜在促肿瘤和感染载体风险。机器灌注研究则常受限于样本量小、器官

来源不均一和灌注参数差异大，跨研究比较并不容易。

还有一点容易被低估：EV 可能同时是治疗物和生物标志物。如果治疗性 EV 改变了灌注液 EV 谱，后续以 EV 作为质量读数时必须区分内源性释放和外源性给药残留。未来研究需要把可追踪标签、单囊泡分析和组织空间组学纳入方案，否则很难证明治疗性 EV 到达了正确细胞并产生了预期效应。

质控和监管是从动物实验走向临床的核心瓶颈。EV 作为治疗产品时，最低限度应报告来源细胞身份、培养体系、收获条件、分离纯化方法、粒径分布、颗粒/蛋白比、典型阳性和阴性 marker、残留 DNA/宿主细胞蛋白、内毒素、无菌、支原体、促凝活性和储存稳定性。更重要的是效价试验必须与适应症相连：用于肾 IRI 时，效价可围绕肾小管上皮损伤、补体沉积或内皮代谢恢复；用于肝移植时，应纳入胆管上皮、Kupffer 细胞和脂肪变性背景；用于心脏 DCD 时，应更强调线粒体结构、收缩功能和心肌酶释放。没有器官和机制特异性的效价，批间一致性就只是物理颗粒的一致，而不是药效的一致。

另一个现实问题是灌注平台本身也在快速变化。连续 NMP、末端 NMP、HOPE、双低温含氧灌注、NRP 后离体灌注、控制性复温和移动式心肺灌注设备的临床目标并不相同 [43-45]。如果 EV 试验不限定平台和场景，就会把物流优化、供体选择、灌注液配方、氧合策略和 EV 药效混在一起。早期研究更适合选择高风险但机制明确的场景，例如 DCD 肾、脂肪变性肝、边缘 DCD 心脏或长冷缺血后肺灌注，并在同一平台内比较 EV 给药与安慰剂，而不是跨器官、跨平台寻找统一结论。

临床试验设计应采用“安全性-机制药效-临床获益”分层。第一阶段可在被弃用或研究用器官中验证 EV 分布、洗脱、促凝风险和灌注参数安全性；第二阶段在可移植高风险器官中采用随机、盲法、灌注平台分层设计，以灌注液 EV/miRNA、补体、组织损伤、dd-cfDNA 和早期功能为共同端点；第三阶段再评价 DGF、PGD、胆道并发症、急性排斥和一年移植物功能。由于边缘供体异质性大，传统大样本固定设计可能成本高且推进慢；小样本自适应、贝叶斯连续监测和中心内平台一致化，更适合用于确定剂量、窗口和富集人群 [8, 35, 46]。

6 展望：未来 3-5 年研究方向

1. 建立“灌注液 EV 药代学”。在 NMP、HOPE 和控制性复温中系统测定 EV 的半衰期、内皮摄取、组织分布、洗脱效率和 cargo 保留，以器官为单位建立剂量-暴露-效应曲线。
2. 发展可追踪、可放行的 EV 质控体系。建议把粒径/浓度、污染蛋白、关键 miRNA/蛋白 cargo、促凝风险、内毒素、无菌、效价模型和批间一致性作为最小放行集。
3. 将 EV 递送与补体/线粒体靶向联合。单一 EV 抗炎效应可能不足以压制 IRI 网络；更现实的策略是在灌注期联合 EV、补体抑制、线粒体保护和内皮糖萼保护，并用灌注液读数实时判断是否达到生物学效应 [2, 10]。
4. 构建“灌注液-组织-外周血”三层预测模型。灌注液 EV/miRNA、代谢组和蛋白组负责移植前风险；植入后 dd-cfDNA 和炎症指标负责早期损伤；病理和长期功能负责模型校准。
5. 设计小样本自适应临床前-临床桥接研究。高风险 DCD 或扩展标准供体器官最适合作为早期场景，端点应优先选择安全性、器官利用率、功能延迟、早期 dd-cfDNA 和组织损伤，而

不是过早追求长期生存差异。

6. 把灌注模式纳入适应证标签。同一种 EV 产品在 HOPE、NMP 和控制性复温中的暴露、摄取和风险不同，未来试验应把温度、氧合、灌注液、给药时点和洗脱策略写入方案，而不是笼统描述为“机器灌注给药”。
7. 发展内源性 EV 与外源性 EV 的区分工具。可追踪膜标签、物种特异核酸、单囊泡蛋白组、空间转录组和组织成像应成为关键临床前证据，用来证明治疗性 EV 的靶细胞、停留时间和下游效应。

7 结论

EV 与机器灌注的交汇，为移植 IRI 研究提供了更可操作的转化场景。机器灌注提供时间、空间和监测窗口；EV 提供可塑的信号载体、治疗载体和机制探针。短期内，灌注液 EV/miRNA 作为器官质量读数，可能比 EV 治疗更快进入临床决策，因为它直接服务于器官取舍、风险分层和术后监测。中期看，标准化 EV 产品与灌注期联合干预有望成为边缘供体器官修复的重要方向，但前提是明确 EV 来源、cargo、效价、给药窗口、器官内分布和长期免疫后果。这个领域能否走得稳，取决于能否把抗炎叙事转化为可计量、可重复、可监管，并且真正改变移植决策的证据链；也取决于研究者能否把 HOPE、NMP 和控制性复温视为不同药理环境，而不是同一个“离体平台”的不同名称。

References

- [1] LIU J, MAN K. Mechanistic insight and clinical implications of ischemia/reperfusion injury post liver transplantation[J/OL]. Cellular and Molecular Gastroenterology and Hepatology, 2023. <https://openalex.org/W4327921881>. DOI: 10.1016/j.jcmgh.2023.03.003.
- [2] CHRISTOU C D, JARADAT F, OLSBURGH J, et al. Rationale for targeting complement to mitigate renal transplant ischemia-reperfusion injury[J/OL]. Journal of the American Society of Nephrology, 2025. <https://openalex.org/W4412559190>. DOI: 10.1681/asn.0000000826.
- [3] MATHIS S, PUTZER G, SCHNEEBERGER S, et al. The endothelial glycocalyx and organ preservation—from physiology to possible clinical implications for solid organ transplantation[J/OL]. International Journal of Molecular Sciences, 2021. <https://openalex.org/W3155926886>. DOI: 10.3390/ijms22084019.
- [4] ZHANG M, LIU Q, MENG H, et al. Ischemia-reperfusion injury: molecular mechanisms and therapeutic targets[J/OL]. Signal Transduction and Targeted Therapy, 2024. <https://openalex.org/W4390655934>. DOI: 10.1038/s41392-023-01688-x.
- [5] ŽULPAITĖ R, MIKNEVICIUS P, LEBER B, et al. Ex-vivo kidney machine perfusion: Therapeutic potential [J/OL]. Frontiers in Medicine, 2021. <https://openalex.org/W4200175990>. DOI: 10.3389/fmed.2021.808719.
- [6] FOGUENNE M, MACMILLAN S, KRON P, et al. Current evidence and future perspectives to implement continuous and end-ischemic use of normothermic and oxygenated hypothermic machine perfusion in clinical practice[J/OL]. Journal of Clinical Medicine, 2023. <https://openalex.org/W4367598714>. DOI: 10.3390/jcm12093207.
- [7] HASJIM B J, SANDERS J M, ALEXANDER M P, et al. Perfusion techniques in kidney allograft preservation

- to reduce ischemic reperfusion injury: A systematic review and meta-analysis[J/OL]. *Antioxidants*, 2024. <https://openalex.org/W4398775947>. DOI: 10.3390/antiox13060642.
- [8] SPIERS H, STADLER L K J, SMITH H, et al. Extracellular vesicles as drug delivery systems in organ transplantation: The next frontier[J/OL]. *Pharmaceutics*, 2023. <https://openalex.org/W4323850442>. DOI: 10.3390/pharmaceutics15030891.
 - [9] GILBO N, BLONDEEL J, PIRENNE J, et al. Organ repair and regeneration during ex situ dynamic preservation: The future is nano[J/OL]. *Transplant International*, 2023. <https://openalex.org/W4388572455>. DOI: 10.3389/ti.2023.11947.
 - [10] PAN W, LI S, LI K, et al. Mesenchymal stem cells and extracellular vesicles: Therapeutic potential in organ transplantation[J/OL]. *Stem Cells International*, 2024. <https://openalex.org/W4395666652>. DOI: 10.1155/2024/2043550.
 - [11] CHEN Y, LI Z, ZHANG H, et al. Mitochondrial metabolism and targeted treatment strategies in ischemic-induced acute kidney injury[J/OL]. *Cell Death Discovery*, 2024. <https://openalex.org/W4391723083>. DOI: 10.1038/s41420-024-01843-5.
 - [12] CHULLO G, PANISELLO-ROSELLÓ A, MARQUEZ N, et al. Focusing on ischemic reperfusion injury in the new era of dynamic machine perfusion in liver transplantation[J/OL]. *International Journal of Molecular Sciences*, 2024. <https://openalex.org/W4390949428>. DOI: 10.3390/ijms25021117.
 - [13] DUGBARTEY G J. Cellular and molecular mechanisms of cell damage and cell death in ischemia-reperfusion injury in organ transplantation[J/OL]. *Molecular Biology Reports*, 2024. <https://openalex.org/W4393305409>. DOI: 10.1007/s11033-024-09261-7.
 - [14] OVEILI E, VAFAEI S, BAZAVAR H, et al. The potential use of mesenchymal stem cells-derived exosomes as micrnas delivery systems in different diseases[J/OL]. *Cell Communication and Signaling*, 2023. <https://openalex.org/W4317734954>. DOI: 10.1186/s12964-022-01017-9.
 - [15] ZHAO B, WEI J, JIANG Z, et al. Mesenchymal stem cell-derived exosomes: an emerging therapeutic strategy for hepatic ischemia-reperfusion injury[J/OL]. *Stem Cell Research & Therapy*, 2025. <https://openalex.org/W4409410618>. DOI: 10.1186/s13287-025-04302-9.
 - [16] GE X, MENG Q, Lü W, et al. Myocardial ischemia-reperfusion induced cardiac extracellular vesicles harbour proinflammatory features and aggravate heart injury[J/OL]. *Journal of Extracellular Vesicles*, 2021. <https://openalex.org/W3130773856>. DOI: 10.1002/jev2.12072.
 - [17] HULLEGIE-PEELEN D M, HESSELINK D A, DIETERICH M, et al. Tissue-resident lymphocytes are released during hypothermic and normothermic machine perfusion of human donor kidneys[J/OL]. *Transplantation*, 2024. <https://openalex.org/W4393385925>. DOI: 10.1097/tp.0000000000004936.
 - [18] CHANCHAROENTHANA W, TRAITANON O, LEELAHAVANICHKUL A, et al. Molecular immune monitoring in kidney transplant rejection: a state-of-the-art review[J/OL]. *Frontiers in Immunology*, 2023. <https://openalex.org/W4386069172>. DOI: 10.3389/fimmu.2023.1206929.
 - [19] RADAJEWSKA A, KRZYWONOS-ZAWADZKA A, BIL-LULA I. Recent methods of kidney storage and therapeutic possibilities of transplant kidney[J/OL]. *Biomedicines*, 2022. <https://openalex.org/W4282931193>. DOI: 10.3390/biomedicines10051013.
 - [20] SHEN C, CHENG H, ZONG T, et al. The role of normothermic machine perfusion (nmp) in the preservation of ex-vivo liver before transplantation: A review[J/OL]. *Frontiers in Bioengineering and Biotechnology*, 2023. <https://openalex.org/W4319758712>. DOI: 10.3389/fbioe.2023.1072937.

- [21] LI J, PENG Q, YANG R, et al. Application of mesenchymal stem cells during machine perfusion: An emerging novel strategy for organ preservation[J/OL]. *Frontiers in Immunology*, 2021. <https://openalex.org/W4200265887>. DOI: [10.3389/fimmu.2021.713920](https://doi.org/10.3389/fimmu.2021.713920).
- [22] RAMPINO T, GREGORINI M, GERMINARIO G, et al. Extracellular vesicles derived from mesenchymal stromal cells delivered during hypothermic oxygenated machine perfusion repair ischemic/reperfusion damage of kidneys from extended criteria donors[J/OL]. *Biology*, 2022. <https://openalex.org/W4213425139>. DOI: [10.3390/biology11030350](https://doi.org/10.3390/biology11030350).
- [23] BURDEYRON P, GIRAUD S, LEPOITTEVIN M, et al. Dynamic conditioning of porcine kidney grafts with extracellular vesicles derived from urine progenitor cells: A proof-of-concept study[J/OL]. *Clinical and Translational Medicine*, 2024. <https://openalex.org/W4405399289>. DOI: [10.1002/ctm2.70095](https://doi.org/10.1002/ctm2.70095).
- [24] TOLOMEO A M, MALVICINI R, VENTRELLA D, et al. Protective effects of mesenchymal stem cells-derived extracellular vesicles against ischemia-reperfusion injury of hearts donated after circulatory death: Preliminary study in a pig model[J/OL]. *Biomedicine & Pharmacotherapy*, 2024. <https://openalex.org/W4401388520>. DOI: [10.1016/j.biopha.2024.117256](https://doi.org/10.1016/j.biopha.2024.117256).
- [25] SUN Y, LIU G, ZHANG K, et al. Mesenchymal stem cells-derived exosomes for drug delivery[J/OL]. *Stem Cell Research & Therapy*, 2021. <https://openalex.org/W3209376853>. DOI: [10.1186/s13287-021-02629-7](https://doi.org/10.1186/s13287-021-02629-7).
- [26] HAN Y, YANG J, FANG J, et al. The secretion profile of mesenchymal stem cells and potential applications in treating human diseases[J/OL]. *Signal Transduction and Targeted Therapy*, 2022. <https://openalex.org/W4220659756>. DOI: [10.1038/s41392-022-00932-0](https://doi.org/10.1038/s41392-022-00932-0).
- [27] TAN F, LI X, WANG Z, et al. Clinical applications of stem cell-derived exosomes[J/OL]. *Signal Transduction and Targeted Therapy*, 2024. <https://openalex.org/W4390765704>. DOI: [10.1038/s41392-023-01704-0](https://doi.org/10.1038/s41392-023-01704-0).
- [28] LI X, WU L, TIAN X, et al. mir-29a-3p in exosomes from heme oxygenase-1 modified bone marrow mesenchymal stem cells alleviates steatotic liver ischemia-reperfusion injury in rats by suppressing ferroptosis via iron responsive element binding protein 2[J/OL]. *Oxidative Medicine and Cellular Longevity*, 2022. <https://openalex.org/W4285740608>. DOI: [10.1155/2022/6520789](https://doi.org/10.1155/2022/6520789).
- [29] WU L, TIAN X, ZUO H, et al. mir-124-3p delivered by exosomes from heme oxygenase-1 modified bone marrow mesenchymal stem cells inhibits ferroptosis to attenuate ischemia-reperfusion injury in steatotic grafts[J/OL]. *Journal of Nanobiotechnology*, 2022. <https://openalex.org/W4224273882>. DOI: [10.1186/s12951-022-01407-8](https://doi.org/10.1186/s12951-022-01407-8).
- [30] LIN F, CHEN W, ZHOU J, et al. Mesenchymal stem cells protect against ferroptosis via exosome-mediated stabilization of slc7a11 in acute liver injury[J/OL]. *Cell Death and Disease*, 2022. <https://openalex.org/W4220817059>. DOI: [10.1038/s41419-022-04708-w](https://doi.org/10.1038/s41419-022-04708-w).
- [31] LUO Y, GUO J, ZHANG P, et al. Mesenchymal stem cell protects injured renal tubular epithelial cells by regulating mtor-mediated th17/treg axis[J/OL]. *Frontiers in Immunology*, 2021. <https://openalex.org/W3170891599>. DOI: [10.3389/fimmu.2021.684197](https://doi.org/10.3389/fimmu.2021.684197).
- [32] STIMMEDER S, LEBER B, SUCHER R, et al. Genetic modulation: Future trends toward graft optimization during machine perfusion[J/OL]. *Transplantation*, 2023. <https://openalex.org/W4385186972>. DOI: [10.1097/tp.0000000000004738](https://doi.org/10.1097/tp.0000000000004738).
- [33] FU Y, WANG Y, LIANG L, et al. Mesenchymal stem cell utilization for in vitro donor liver machine perfusion preservation: Current status and future directions[J/OL]. *Stem Cells Translational Medicine*, 2023. <https://openalex.org/W4386243629>. DOI: [10.1093/stcltm/szad053](https://doi.org/10.1093/stcltm/szad053).

- [34] CILLO U, LONATI C, BERTACCO A, et al. A proof-of-concept study in small and large animal models for coupling liver normothermic machine perfusion with mesenchymal stromal cell bioreactors[J/OL]. *Nature Communications*, 2025. <https://openalex.org/W4406016891>. DOI: 10.1038/s41467-024-55217-7.
- [35] BOGENSPERGER C, HOFMANN J, MESSNER F, et al. Ex vivo mesenchymal stem cell therapy to regenerate machine perfused organs[J/OL]. *International Journal of Molecular Sciences*, 2021. <https://openalex.org/W3163658231>. DOI: 10.3390/ijms22105233.
- [36] DE NECOCHEA CAMPION R, PESQUEIRA M, VALLEJOS P, et al. A lectin affinity plasmapheresis device removes extracellular vesicles and micrnas from renal perfusates following controlled oxygenated rewarming of discarded donor kidneys[J/OL]. *Transplant Immunology*, 2025. <https://openalex.org/W4408050599>. DOI: 10.1016/j.trim.2025.102215.
- [37] JIANG M, ZHENG Z, WANG X, et al. A biomarker framework for liver aging: the aging biomarker consortium consensus statement[J/OL]. *Life Medicine*, 2024. <https://openalex.org/W4391673165>. DOI: 10.1093/lifemedi/lnae004.
- [38] HUANG C, SU P, FISHER T D, et al. Mass spectrometry-based proteomics for advancing solid organ transplantation research[J/OL]. *Frontiers in Transplantation*, 2023. <https://openalex.org/W4388976812>. DOI: 10.3389/frtra.2023.1286881.
- [39] DE NECOCHEA CAMPION R, PESQUEIRA M, VALLEJOS P, et al. The hemopurifier removes extracellular vesicles and micrnas from renal perfusates following controlled oxygenated rewarming of discarded donor kidneys[J/OL]. *bioRxiv (Cold Spring Harbor Laboratory)*, 2024. <https://openalex.org/W4401856794>. DOI: 10.1101/2024.08.23.609252.
- [40] BRONKHORST A J, UNGERER V, OBERHOFER A, et al. New perspectives on the importance of cell-free dna biology[J/OL]. *Diagnostics*, 2022. <https://openalex.org/W4294792883>. DOI: 10.3390/diagnostics12092147.
- [41] DOUVRIIS A, VIÑAS J L, AKBARI S, et al. Systematic review of micrnas in human acute kidney injury [J/OL]. *Renal Failure*, 2024. <https://openalex.org/W4403963215>. DOI: 10.1080/0886022x.2024.2419960.
- [42] QIU S, CAI Y, YAO H, et al. Small molecule metabolites: discovery of biomarkers and therapeutic targets [J/OL]. *Signal Transduction and Targeted Therapy*, 2023. <https://openalex.org/W4327947056>. DOI: 10.1038/s41392-023-01399-3.
- [43] BOWLES D E, KARIMOV J H, AMARELLI C. Editorial: Graft preservation[J/OL]. *Frontiers in Cardiovascular Medicine*, 2024. <https://openalex.org/W4403747999>. DOI: 10.3389/fcvm.2024.1478730.
- [44] UGHETTO A, ROUBILLE F, MOLINA A, et al. Heart graft preservation technics and limits: an update and perspectives[J/OL]. *Frontiers in Cardiovascular Medicine*, 2023. <https://openalex.org/W4388414151>. DOI: 10.3389/fcvm.2023.1248606.
- [45] VON HORN C, LüER B, MALKUS L, et al. Comparison between terminal or preterminal conditioning of donor livers by ex situ machine perfusion[J/OL]. *Transplantation*, 2023. <https://openalex.org/W4327614582>. DOI: 10.1097/tp.0000000000004568.
- [46] FRANZIN R, STASI A, FIORENTINO M, et al. Renal delivery of pharmacologic agents during machine perfusion to prevent ischaemia-reperfusion injury: From murine model to clinical trials[J/OL]. *Frontiers in Immunology*, 2021. <https://openalex.org/W3180583737>. DOI: 10.3389/fimmu.2021.673562.
- [47] HE J, KHAN U Z, QING L, et al. Improving the ischemia-reperfusion injury in vascularized composite

- allotransplantation: Clinical experience and experimental implications[J/OL]. *Frontiers in Immunology*, 2022. <https://openalex.org/W4296049160>. DOI: 10.3389/fimmu.2022.998952.
- [48] CHEN H, LU D, YANG X, et al. One shoot, two birds: Alleviating inflammation caused by ischemia/reperfusion injury to reduce the recurrence of hepatocellular carcinoma[J/OL]. *Frontiers in Immunology*, 2022. <https://openalex.org/W4280498796>. DOI: 10.3389/fimmu.2022.879552.
- [49] AKALAY S, HOSGOOD S A. How to best protect kidneys for transplantation—mechanistic target[J/OL]. *Journal of Clinical Medicine*, 2023. <https://openalex.org/W4321597205>. DOI: 10.3390/jcm12051787.
- [50] PATRONO D, STEFANO N D, VISSIO E, et al. How to preserve steatotic liver grafts for transplantation [J/OL]. *Journal of Clinical Medicine*, 2023. <https://openalex.org/W4380365344>. DOI: 10.3390/jcm12123982.
- [51] STORTI G, FAVI E, ALBANESI F, et al. Adipose-derived stem/stromal cells in kidney transplantation: Status quo and future perspectives[J/OL]. *International Journal of Molecular Sciences*, 2021. <https://openalex.org/W3205131525>. DOI: 10.3390/ijms222011188.
- [52] FRANZIN R, STASI A, RANIERI E, et al. Targeting premature renal aging: from molecular mechanisms of cellular senescence to senolytic trials[J/OL]. *Frontiers in Pharmacology*, 2021. <https://openalex.org/W3159424329>. DOI: 10.3389/fphar.2021.630419.
- [53] SHAO B, QIN Y, REN S, et al. Structural and temporal dynamics of mesenchymal stem cells in liver diseases from 2001 to 2021: A bibliometric analysis[J/OL]. *Frontiers in Immunology*, 2022. <https://openalex.org/W4280528934>. DOI: 10.3389/fimmu.2022.859972.
- [54] CUI Z, ZHAO X, ZHAO X, et al. Therapeutic application of quercetin in aging-related diseases: Sirt1 as a potential mechanism[J/OL]. *Frontiers in Immunology*, 2022. <https://openalex.org/W4286447743>. DOI: 10.3389/fimmu.2022.943321.
- [55] PALUSZKIEWICZ P, MARTUSZEWSKI A, ZARĘBA N, et al. The application of nanoparticles in diagnosis and treatment of kidney diseases[J/OL]. *International Journal of Molecular Sciences*, 2021. <https://openalex.org/W4200069478>. DOI: 10.3390/ijms23010131.
- [56] PATEL P M, CONNOLLY M R, COE T M, et al. Minimizing ischemia reperfusion injury in xenotransplantation[J/OL]. *Frontiers in Immunology*, 2021. <https://openalex.org/W3196561138>. DOI: 10.3389/fimmu.2021.681504.
- [57] . ., . ., ZORKINA Y, . Human Gut Microbiota for Diagnosis and Treatment of Depression [J/OL]. *International Journal of Molecular Sciences*, 2024. <https://openalex.org/W4399055080>. DOI: 10.3390/ijms25115782.
- [58] HUANG R, FU P, MA L. Kidney fibrosis: from mechanisms to therapeutic medicines[J/OL]. *Signal Transduction and Targeted Therapy*, 2023. <https://openalex.org/W4327811820>. DOI: 10.1038/s41392-023-01379-7.
- [59] MELNIK B C, STREMMEL W, WEISKIRCHEN R, et al. Exosome-derived micrornas of human milk and their effects on infant health and development[J/OL]. *Biomolecules*, 2021. <https://openalex.org/W3172470733>. DOI: 10.3390/biom11060851.
- [60] PATEL J C, SHUKLA M, SHUKLA M, et al. From bench to bedside: translating mesenchymal stem cell therapies through preclinical and clinical evidence[J/OL]. *Frontiers in Bioengineering and Biotechnology*, 2025. <https://openalex.org/W4412760870>. DOI: 10.3389/fbioe.2025.1639439.
- [61] VELIKIĆ G, MARIĆ D, MARIĆ D, et al. Harnessing the stem cell niche in regenerative medicine: Innovative

- avenue to combat neurodegenerative diseases[J/OL]. *International Journal of Molecular Sciences*, 2024. <https://openalex.org/W4390818132>. DOI: 10.3390/ijms25020993.
- [62] PAPADOPOULOS K, PIPERI C, KORKOLOPOULOU P. Clinical applications of adipose-derived stem cell (adsc) exosomes in tissue regeneration[J/OL]. *International Journal of Molecular Sciences*, 2024. <https://openalex.org/W4399138458>. DOI: 10.3390/ijms25115916.
- [63] HAUPT M, GERNER S T, BÄHR M, et al. Neuroprotective strategies for ischemic stroke—future perspectives[J/OL]. *International Journal of Molecular Sciences*, 2023. <https://openalex.org/W4321497223>. DOI: 10.3390/ijms24054334.
- [64] BARRÈRE-LEMAIRE S, VINCENT A, JØRGENSEN C, et al. Mesenchymal stromal cells for improvement of cardiac function following acute myocardial infarction: a matter of timing[J/OL]. *Physiological Reviews*, 2023. <https://openalex.org/W4385898210>. DOI: 10.1152/physrev.00009.2023.
- [65] ZARIPOVA L, MIDGLEY A, CHRISTMAS S E, et al. Mesenchymal stem cells in the pathogenesis and therapy of autoimmune and autoinflammatory diseases[J/OL]. *International Journal of Molecular Sciences*, 2023. <https://openalex.org/W4388464826>. DOI: 10.3390/ijms242216040.
- [66] MICELI V, BULATI M, GALLO A, et al. Role of mesenchymal stem/stromal cells in modulating ischemia/reperfusion injury: Current state of the art and future perspectives[J/OL]. *Biomedicines*, 2023. <https://openalex.org/W4321790344>. DOI: 10.3390/biomedicines11030689.
- [67] LOTFY A, ABOQUELLA N M, WANG H. Mesenchymal stromal/stem cell (msc)-derived exosomes in clinical trials[J/OL]. *Stem Cell Research & Therapy*, 2023. <https://openalex.org/W4362659077>. DOI: 10.1186/s13287-023-03287-7.
- [68] GUO J, HUANG X, DOU L, et al. Aging and aging-related diseases: from molecular mechanisms to interventions and treatments[J/OL]. *Signal Transduction and Targeted Therapy*, 2022. <https://openalex.org/W4311744984>. DOI: 10.1038/s41392-022-01251-0.
- [69] NOWAK-STĘPNIOWSKA A, OSUCHOWSKA P N, FIEDOROWICZ H, et al. Insight in hypoxia-mimetic agents as potential tools for mesenchymal stem cell priming in regenerative medicine[J/OL]. *Stem Cells International*, 2022. <https://openalex.org/W4221045129>. DOI: 10.1155/2022/8775591.
- [70] WANG S, LEI B, ZHANG E, et al. Targeted therapy for inflammatory diseases with mesenchymal stem cells and their derived exosomes: From basic to clinics[J/OL]. *International Journal of Nanomedicine*, 2022. <https://openalex.org/W4224280803>. DOI: 10.2147/ijn.s355366.
- [71] CAO Q, HUANG C, CHEN X, et al. Mesenchymal stem cell-derived exosomes: Toward cell-free therapeutic strategies in chronic kidney disease[J/OL]. *Frontiers in Medicine*, 2022. <https://openalex.org/W4220722972>. DOI: 10.3389/fmed.2022.816656.
- [72] LIN Z, WU Y, XU Y, et al. Mesenchymal stem cell-derived exosomes in cancer therapy resistance: recent advances and therapeutic potential[J/OL]. *Molecular Cancer*, 2022. <https://openalex.org/W4295750964>. DOI: 10.1186/s12943-022-01650-5.
- [73] MARGIANA R, MARKOV A, ZEKIY A O, et al. Clinical application of mesenchymal stem cell in regenerative medicine: a narrative review[J/OL]. *Stem Cell Research & Therapy*, 2022. <https://openalex.org/W4288084885>. DOI: 10.1186/s13287-022-03054-0.
- [74] MARKOV A, THANGAVELU L, SURENDAR A, et al. Retracted article: Mesenchymal stem/stromal cells as a valuable source for the treatment of immune-mediated disorders[J/OL]. *Stem Cell Research & Therapy*, 2021. <https://openalex.org/W3138150943>. DOI: 10.1186/s13287-021-02265-1.

- [75] HASSANZADEH A, RAHMAN H S, MARKOV A, et al. Mesenchymal stem/stromal cell-derived exosomes in regenerative medicine and cancer; overview of development, challenges, and opportunities[J/OL]. Stem Cell Research & Therapy, 2021. <https://openalex.org/W3161946478>. DOI: 10.1186/s13287-021-02378-7.
- [76] MOGHADASI S, ELVENY M, RAHMAN H S, et al. Retracted article: A paradigm shift in cell-free approach: the emerging role of mscs-derived exosomes in regenerative medicine[J/OL]. Journal of Translational Medicine, 2021. <https://openalex.org/W3178800228>. DOI: 10.1186/s12967-021-02980-6.
- [77] CAO H, WU L, TIAN X, et al. Ho-1/bmmsc perfusion using a normothermic machine perfusion system reduces the acute rejection of dcd liver transplantation by regulating nkt cell co-inhibitory receptors in rats[J/OL]. Stem Cell Research & Therapy, 2021. <https://openalex.org/W3152628611>. DOI: 10.1186/s13287-021-02647-5.
- [78] BRUNO S, SANCHEZ M B H, CHIABOTTO G, et al. Human liver stem cells: A liver-derived mesenchymal stromal cell-like population with pro-regenerative properties[J/OL]. Frontiers in Cell and Developmental Biology, 2021. <https://openalex.org/W3161083643>. DOI: 10.3389/fcell.2021.644088.
- [79] HADE M D, SUIRE C N, SUO Z. Mesenchymal stem cell-derived exosomes: Applications in regenerative medicine[J/OL]. Cells, 2021. <https://openalex.org/W3188331302>. DOI: 10.3390/cells10081959.
- [80] DABROWSKA S, ANDRZEJEWSKA A, JANOWSKI M, et al. Immunomodulatory and regenerative effects of mesenchymal stem cells and extracellular vesicles: Therapeutic outlook for inflammatory and degenerative diseases[J/OL]. Frontiers in Immunology, 2021. <https://openalex.org/W3127151176>. DOI: 10.3389/fimmu.2020.591065.
- [81] JANOČKOVÁ J, SLOVINSKÁ L, HARVANOVÁ D, et al. New therapeutic approaches of mesenchymal stem cells-derived exosomes[J/OL]. Journal of Biomedical Science, 2021. <https://openalex.org/W3165520006>. DOI: 10.1186/s12929-021-00736-4.
- [82] GUILLAMAT-PRATS R. The role of msc in wound healing, scarring and regeneration[J/OL]. Cells, 2021. <https://openalex.org/W3182042652>. DOI: 10.3390/cells10071729.
- [83] ZHU D, LI Z, HUANG K, et al. Minimally invasive delivery of therapeutic agents by hydrogel injection into the pericardial cavity for cardiac repair[J/OL]. Nature Communications, 2021. <https://openalex.org/W3133794548>. DOI: 10.1038/s41467-021-21682-7.
- [84] TRZYNA A, BANAŚ-ZĄBCZYK A. Adipose-derived stem cells secretome and its potential application in “stem cell-free therapy”[J/OL]. Biomolecules, 2021. <https://openalex.org/W3168570978>. DOI: 10.3390/biom11060878.
- [85] SHARMA V, DASH S K, GOVARTHANAN K, et al. Recent advances in cardiac tissue engineering for the management of myocardium infarction[J/OL]. Cells, 2021. <https://openalex.org/W3203880271>. DOI: 10.3390/cells10102538.
- [86] HUANG Y, YANG L. Mesenchymal stem cells and extracellular vesicles in therapy against kidney diseases[J/OL]. Stem Cell Research & Therapy, 2021. <https://openalex.org/W3146209640>. DOI: 10.1186/s13287-021-02289-7.
- [87] ZHUANG W Z, LIN Y, SU L, et al. Mesenchymal stem/stromal cell-based therapy: mechanism, systemic safety and biodistribution for precision clinical applications[J/OL]. Journal of Biomedical Science, 2021. <https://openalex.org/W3153331636>. DOI: 10.1186/s12929-021-00725-7.
- [88] XU X, PANG Y, FAN X. Mitochondria in oxidative stress, inflammation and aging: from mechanisms

- to therapeutic advances[J/OL]. Signal Transduction and Targeted Therapy, 2025. <https://openalex.org/W4411192949>. DOI: 10.1038/s41392-025-02253-4.
- [89] HONG W, HUANG H, ZENG X, et al. Targeting mitochondrial quality control: new therapeutic strategies for major diseases[J/OL]. Military Medical Research, 2024. <https://openalex.org/W4401729535>. DOI: 10.1186/s40779-024-00556-1.
- [90] YAO Z, LI H, LIAO P, et al. Mitochondrial dysfunction: mechanisms and advances in therapy[J/OL]. Signal Transduction and Targeted Therapy, 2024. <https://openalex.org/W4396936983>. DOI: 10.1038/s41392-024-01839-8.